

CELL DIVISION

Mitosis

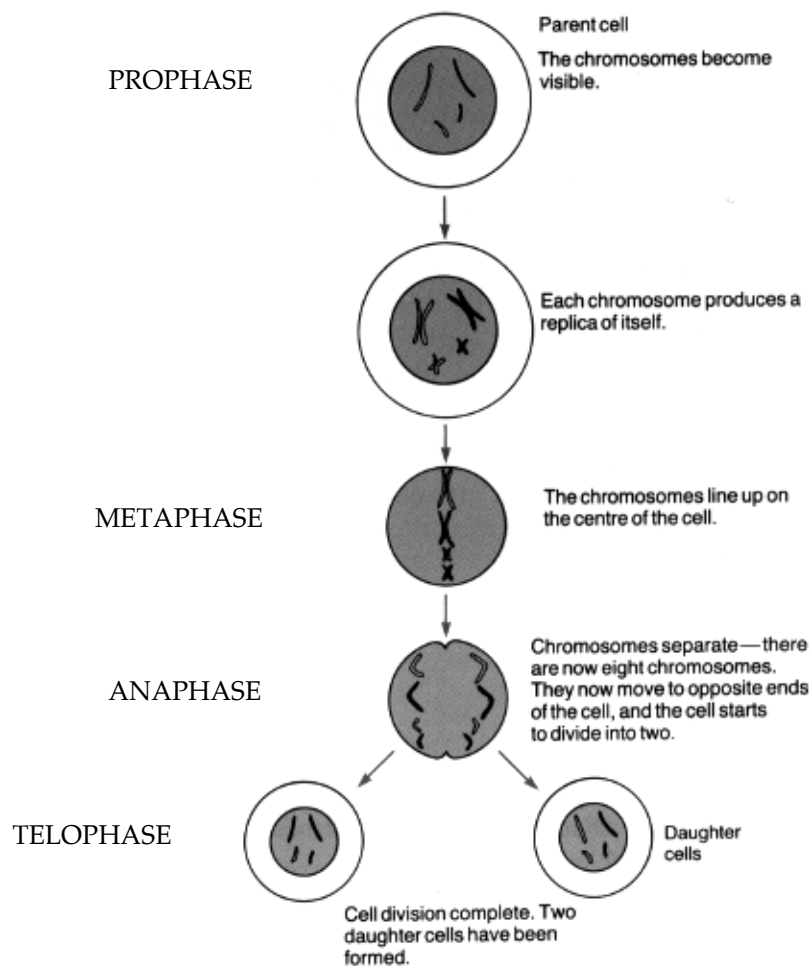
Mitosis produces new cells for *growth* and to *replace damaged cells*.

It occurs for *all cells except for the reproductive cells* (sperm and egg cells).

One cell divides many times and each cell produced will be an *exact copy of the one cell* with *all cells having the same number of chromosomes*.

The nucleus controls the division process.

The steps involved in this process are shown below.



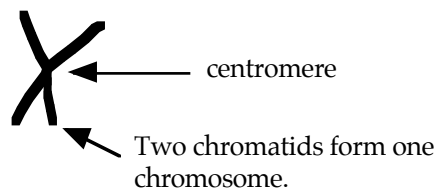
INTERPHASE - chromosomes are not visible.

Note

The terms prophase, metaphase, anaphase, telophase and interphase are *useful for Upper School* only.

When the chromosomes first become visible and then duplicate themselves, they form ***two chromatids*** that are joined by a ***centromere***.

This still ***represents one chromosome only***, not two.



The result of mitosis is that the daughter cells all contain the ***normal number of chromosomes***.

Definitions

Diploid Number - normal number of chromosomes.

Haploid Number - half the normal number of chromosomes.

Meiosis

This occurs in the *reproductive organs* and produces *gametes with the haploid number of chromosomes* (half the normal number).

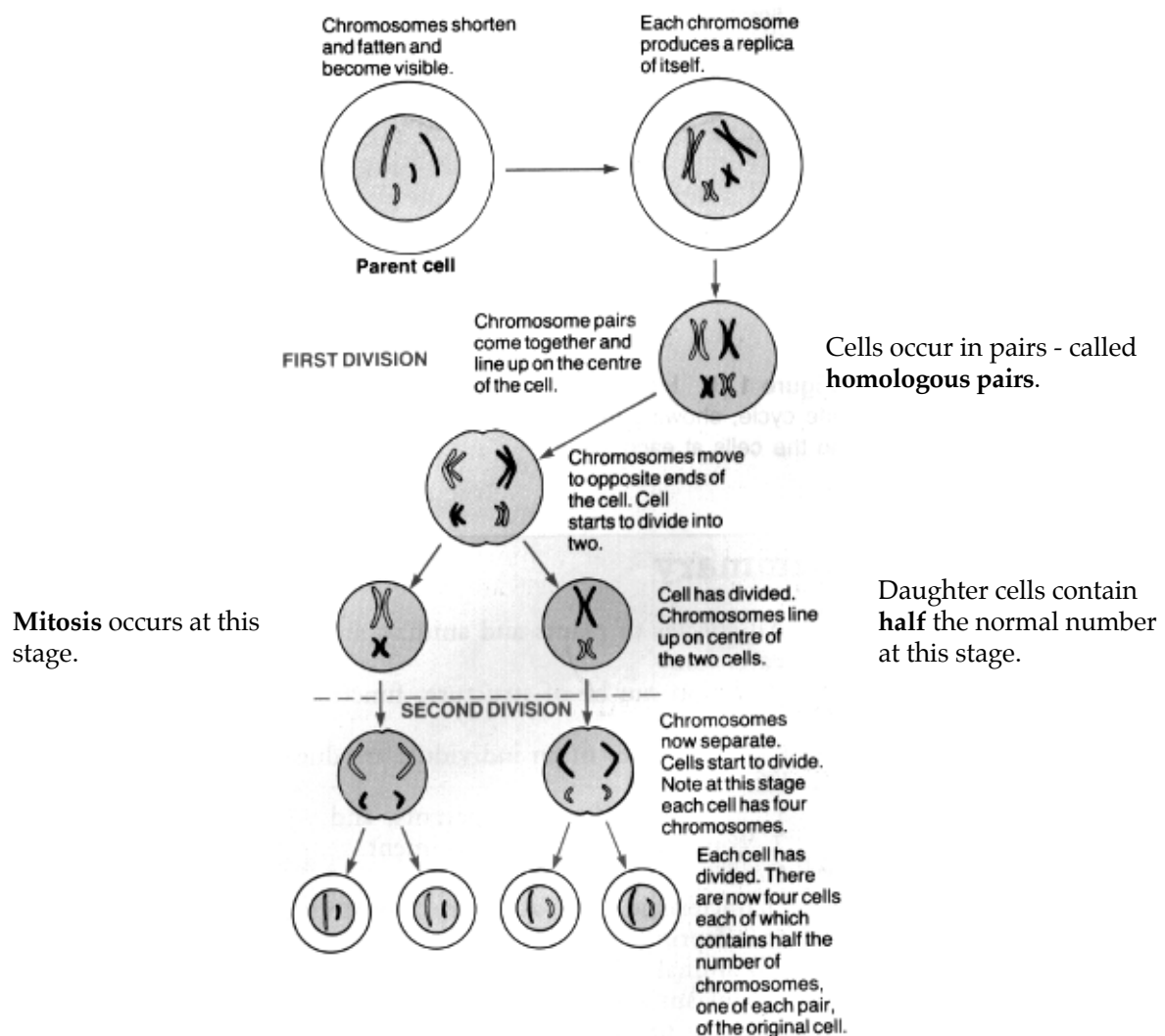
This is important as during the *fertilisation* process between an egg and sperm cell, the *diploid or normal number of chromosomes is restored*.

The process occurs in two stages.

STAGE 1 - the chromosome number is halved.

STAGE 2 - the chromosomes are pulled apart as in mitosis.

The process occurs as shown below.



Definition

Homologous Pair - two chromosomes containing similar genetic information for particular characteristics.